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 Oefeningenreeks 2D nr. 16.6

Los op: $f(x) = \frac{1 + \cos(2x)}{\sin(x)}$ voor: $x \rightarrow \frac{\pi}{2}; x \rightarrow 0; x \rightarrow \pi$

Voor $x \rightarrow \frac{\pi}{2}$: $\lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{1 + \cos(2x)}{\sin(x)} \right) = \frac{1 + \cos(2\frac{\pi}{2})}{\sin(\frac{\pi}{2})} = \frac{1 + -1}{1} = 0$

Voor $x \rightarrow 0$: $\lim_{x \rightarrow 0} \left(\frac{1 + \cos(2x)}{\sin(x)} \right) = \frac{1 + \cos(0)}{\sin(0)} = \frac{1 + 1}{0} = \infty$

Bekijk linker- en rechterlimiet:

$$\lim_{x \rightarrow 0+} \left(\frac{1 + \cos(2x)}{\sin(x)} \right) = \frac{1 + \cos(0+)}{\sin(0+)} = \frac{1 + 1+}{0+} = +\infty$$

$$\lim_{x \rightarrow 0-} \left(\frac{1 + \cos(2x)}{\sin(x)} \right) = \frac{1 + \cos(0-)}{\sin(0-)} = \frac{1 + 1+}{0-} = -\infty$$

Voor $x \rightarrow \pi$: $\lim_{x \rightarrow \pi} \left(\frac{1 + \cos(2x)}{\sin(x)} \right) = \frac{1 + \cos(2\pi)}{\sin(\pi)} = \frac{1 + 1}{0} = \infty$

Bekijk linker- en rechterlimiet:

$$\lim_{x \rightarrow \pi+} \left(\frac{1 + \cos(2x)}{\sin(x)} \right) = \frac{1 + \cos(2\pi+)}{\sin(\pi+)} = \frac{1 + 1+}{0-} = -\infty$$

$$\lim_{x \rightarrow \pi-} \left(\frac{1 + \cos(2x)}{\sin(x)} \right) = \frac{1 + \cos(2\pi-)}{\sin(\pi-)} = \frac{1 + 1+}{0+} = +\infty$$

References